

ITERATIVE METHODS FOR SOLVING LINEAR SYSTEMS

$$A \in \mathbb{R}^{n \times n} \quad b \in \mathbb{R}^n \quad Ax = b \quad x_* := A^{-1}b \quad \text{true solution of the lin. system}$$

Recap of conjugate gradient (CG): A symmetric positive definite

Start with initial guess, for example $x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, and construct

$$x_1, x_2, \dots, x_k, \dots \quad \text{such that} \quad x_j \in K_j(A, b) = \text{span}\{b, Ab, A^2b, \dots, A^{j-1}b\}$$

$$\hookrightarrow x_j \in \underset{x \in K_j(A, b)}{\text{argmin}} \|x - x_*\|_A \rightsquigarrow \|y\|_A = \sqrt{y^T A y}$$

$\hookrightarrow$ Krylov subspace

computational cost to get x_j
 $\approx j$ matrix-vector multiplications with A
 $+ O(nj^2)$ operations

GMRES, MINRES

$\swarrow$
 for any matrix $A \in \mathbb{R}^{n \times n}$

$\hookrightarrow$ for symmetric matrix $A \in \mathbb{R}^{n \times n}$

General idea: look for approximate solutions

$$x_j \in K_j(A, b) \quad \text{for } j = 1, 2, 3, \dots$$

Constructing a ^{orthonormal (nice)} basis of $K_j(A, b) = \text{span}\{b, Ab, A^2b, \dots, A^{j-1}b\}$

A basis of $K_j(A, b)$ is $\underbrace{b, Ab, A^2b, \dots, A^{j-1}b}_{\text{(in general)}}$

$\hookrightarrow$ under suitable assumptions, these vectors (normalized) converge to an eigenvector of A corresponding to the eigenvalue of largest modulus

$\rightsquigarrow$ numerically, they are almost linearly dependent

$\Rightarrow$ ill-conditioned basis $\hat{!}$

$$K_1(A, b) \subseteq K_2(A, b) \subseteq K_3(A, b) \subseteq \dots$$

It will be better to construct an orthonormal basis of the Krylov subspace!

We want vectors $q_1, q_2, \dots, q_k \in \mathbb{R}^n$ with $\|q_i\| = 1$ and $q_i^T q_j = 0$ when $i \neq j$

$$Q_k = \begin{bmatrix} | & | & \dots & | \\ q_1 & q_2 & \dots & q_k \\ | & | & \dots & | \end{bmatrix} \in \mathbb{R}^{n \times k} \quad Q_k^T Q_k = I_k$$

$\{q_1, q_2, \dots, q_j\}$ basis of $K_j(A, b)$ for $j = 1, \dots, k$.

At each step we would like to add a vector q_{j+1} , orthogonal to

all previous vectors, and it is $q_{j+1} \in K_{j+1}(A, b) \setminus K_j(A, b)$

$$\begin{aligned} \text{Def: if } q_j \in K_j(A, b) \quad \text{then} \quad q_j &= \alpha_0 b + \alpha_1 Ab + \alpha_2 A^2 b + \dots + \alpha_{j-1} A^{j-1} b \\ &= (\alpha_0 I + \alpha_1 A + \alpha_2 A^2 + \dots + \alpha_{j-1} A^{j-1}) b \\ &= p(A) \cdot b \end{aligned}$$

for the polynomial $p(t) = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \dots + \alpha_{j-1} t^{j-1}$.

We say that q_j has degree $j-1$ if $\alpha_{j-1} \neq 0$.

We solve the problem with an additional constraint: q_{j+1} has degree j (for all $j=1, \dots, k$).

Vector q_1 in $\mathcal{K}_1(A, b) = \text{span}\{b\} \rightarrow q_1 = b / \|b\|$

Inductively, assume we have $\underbrace{q_1, \dots, q_j}_{\substack{\text{orthonormal,} \\ q_i \in \mathcal{K}_i(A, b), \text{ degree of } q_i \text{ is } i-1}}$, we want to compute q_{j+1}

Let $w = Aq_j \in \mathcal{K}_{j+1}(A, b)$ and has degree j :

we could write $q_j = \alpha_0 b + \alpha_1 A b + \dots + \alpha_{j-1} A^{j-1} b$ $\alpha_{j-1} \neq 0$

$$\Rightarrow w = Aq_j = \alpha_0 Ab + \alpha_1 A^2 b + \dots + \alpha_{j-1} \underbrace{A^j b}_{\Rightarrow w \in \mathcal{K}_{j+1}(A, b) \text{ deg}(w) = j}$$

$$w \in \mathcal{K}_{j+1}(A, b) \setminus \mathcal{K}_j(A, b)$$

Need to make w orthogonal to $q_1, \dots, q_j$

$\Rightarrow$ Subtract its projection onto $q_1, \dots, q_j$ $Q_j = [q_1 \dots q_j]$

for $i=1:j$ $\underbrace{w \leftarrow w - q_i q_i^T w}_{\text{orth. proj}}$
end

$$w \leftarrow w - Q_j(Q_j^T w)$$

Final step: $q_{j+1} \leftarrow w / \|w\|$

Arnoldi algorithm: input: $A \in \mathbb{R}^{n \times n}$ $b \in \mathbb{R}^n$, k

$$q_1 = b / \|b\| \xrightarrow{\text{h}_{21}} \text{ } O(n)$$

for $j=2, \dots, k$

$$w = Aq_{j-1} \quad O(mz(A))$$

for $i=1, \dots, j-1$

$$w = w - q_i \underbrace{(q_i^T w)}_{h_{i,j-1}} \quad O(n) \quad \left. \vphantom{w = w - q_i (q_i^T w)} \right\} O(nj)$$

end

$$q_j = w / \|w\| \quad O(n)$$

end $\xrightarrow{\text{h}_{jj-1}}$

$$\leadsto O(k \cdot mz(A) + nk^2)$$

$$O(n + 2n + 3n + \dots + kn) = O\left(n \cdot \underbrace{(1+2+\dots+k)}_{\frac{k(k+1)}{2}}\right) = O(nk^2)$$

$$Aq_j = q_1 h_{1j} + q_2 h_{2j} + \dots + q_j h_{jj} + q_{j+1} h_{j+1,j}$$

$$Aq_{j-1} = q_1 h_{1,j-1} + q_2 h_{2,j-1} + \dots + q_{j-1} h_{j-1,j-1} + q_j h_{j,j-1}$$

Arnoldi decomposition:

$$A \cdot \begin{bmatrix} q_1 & q_2 & \dots & q_j \end{bmatrix}_{n \times j} = \begin{bmatrix} q_1 & q_2 & \dots & q_{j+1} \end{bmatrix}_{n \times (j+1)} \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1j} \\ h_{21} & h_{22} & \dots & h_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ h_{j1} & h_{j2} & \dots & h_{jj} \\ 0 & 0 & \dots & 0 \end{bmatrix}_{(j+1) \times j}$$

$h_{j+1,j} = h_{j+1,j} e_j^T$

$$A Q_j = Q_{j+1} H_j \rightarrow \text{upper Hessenberg matrix}$$

$$\begin{bmatrix} b_1 & b_k \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} -c_1^T \\ \vdots \\ -c_k^T \end{bmatrix} = b_1 c_1^T + \dots + b_k c_k^T$$

$$\begin{bmatrix} B \\ \vdots \end{bmatrix} \times \begin{bmatrix} C \\ y^T \end{bmatrix} = B C^T + x y^T$$

Questions:

(1) Can the Arnoldi algorithm break down?

(if that happens, what does it mean?)

(2) what happens when A is symmetric?

(what is the shape of the matrix H_j from the Arnoldi decomposition?)

(1) Yes, it can happen that at some point $w = 0$ (at the j -th step)

$\Rightarrow w = \text{linear combination of } q_1, q_2, \dots, q_{j-1}$

$\Rightarrow A \cdot q_{j-1} \in K_{j-1}(A, b) \Rightarrow K_j(A, b) = K_{j-1}(A, b)$

Then $\{q_1, q_2, \dots, q_{j-1}\}$ are also a basis of the j -th Krylov subspace $K_j(A, b)$

(2) $AQ_j = Q_j H_j + h_{j+1,j} q_{j+1} e_j^T$ $H_j = \begin{bmatrix} H_j \\ \vdots & \vdots & h_{j+1,j} \end{bmatrix} \rightarrow H_j \text{ is upper Hessenberg}$

alternative way of writing the Arnoldi decomposition

$$\rightarrow Q_j^T A Q_j = \underbrace{Q_j^T Q_j}_{I_j} H_j + h_{j+1,j} \underbrace{Q_j^T q_{j+1}}_{=0} e_j^T = H_j$$

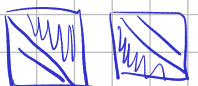
$$= \begin{bmatrix} -q_1^T & \vdots \\ \vdots & -q_j^T \end{bmatrix} = \begin{bmatrix} q_1^T q_{j+1} \\ \vdots \\ q_j^T q_{j+1} \end{bmatrix} = 0$$

$$Q_j^T A Q_j = H_j \quad \text{for all } A \in \mathbb{R}^{n \times n}$$

$$\text{If } A \text{ is symmetric} \quad H_j = Q_j^T A Q_j \Rightarrow$$

$$H_j^T = \underbrace{(Q_j^T A Q_j)^T}_{H_j} = Q_j^T A^T Q_j \underset{\text{symmetry}}{=} Q_j^T A Q_j = H_j$$

$\Rightarrow H_j$ is tridiagonal!



$\Rightarrow$ consequence: at step j , we only need to orthogonalize the new vector $w = Aq_{j-1}$ against q_{j-1} and q_{j-2} .

The inner products with $q_{j-3}, q_{j-4}, \dots, q_1$ are all 0.

$\leadsto$ Lanczos algorithm

inputs: $A \in \mathbb{R}^{n \times n}$ SYMMETRIC, $b \in \mathbb{R}^n$, k

$$q_1 = b / \|b\|$$

for $j = 2 : k$

$$w = A q_{j-1}$$

$$w \leftarrow w - q_{j-1} q_{j-1}^T w$$

if $j \geq 3$

$$w \leftarrow w - q_{j-2} q_{j-2}^T w$$

end

$$q_j = w / \|w\|$$

end

$O(\text{nnz}(A)) \cdot k$ for matrix-vector multiplications

+ $O(nk)$